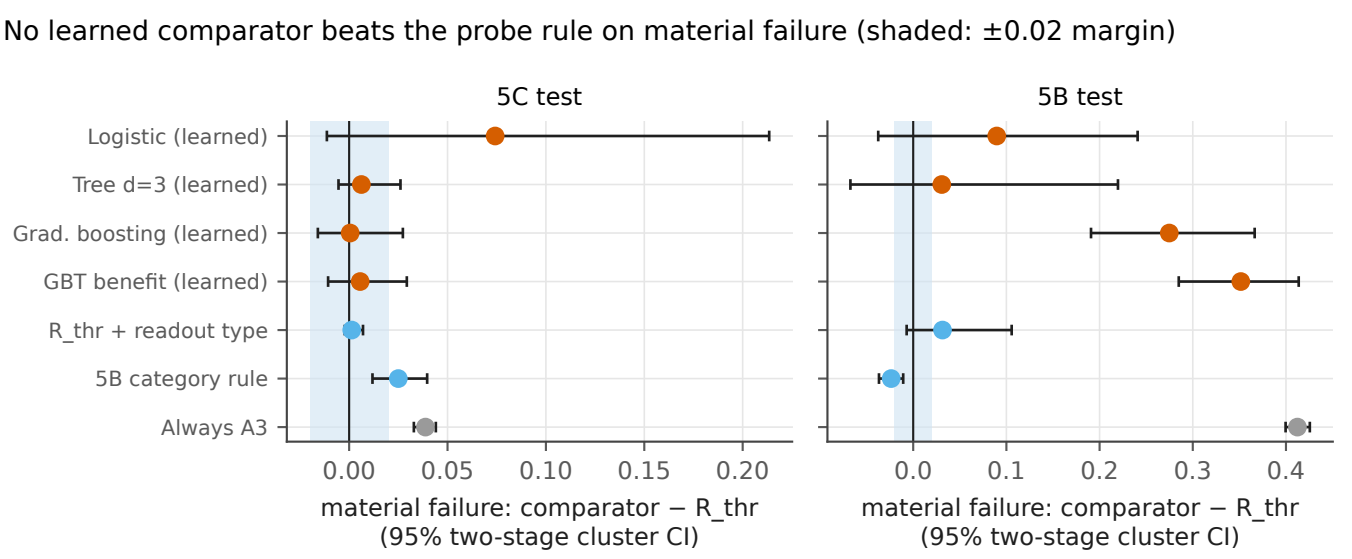
No learned comparator beats the probe rule on material failure (shaded: ±0.02 margin)
5C test
5B test
Logistic (learned)
Tree d=3 (learned)
Grad. boosting (learned)
GBT benefit (learned)
R_thr + readout type
5B category rule
Always A3
0.00
0.05
0.10
0.15
0.20
0.0
0.1
0.2
0.3
0.4
material failure: comparator − R_thr
(95% two-stage cluster CI)
material failure: comparator − R_thr
(95% two-stage cluster CI)